Does a tyre-degradation model transfer?

Cross-championship evidence and a retrospective audit of 1,280 real pit-stop decisions

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Abstract

Race-strategy models are usually fitted and validated inside one championship. Does a fitted model predict an unseen season, and does its recommendation resemble what a team did? I apply one protocol to both across 7 car classes and four championships: Formula 1, WEC, IMSA and ELMS.

Four findings. First, public timing rarely carries enough signal to test transfer. With a slope stable by construction, 18 of 22 Formula 1 circuits cannot reach a within-stint R^2 of 0.2 against their own lap noise; pricing that in leaves 8 unstable and 3 stable, and 5 of 51 endurance circuit-classes reach it. Within-circuit recoding to real Pirelli compounds, Friday long runs and past-only fitting do not help, and a circuit's own slope does not beat one common to the category. Second, in these championships the cost of a stop separates the strategy regimes best: over 205 race-seasons pit loss alone reaches an AUC of 0.977, against 0.865 for degradation. Third, replaying 1,280 real first stops with each team's next stop held fixed, a time-optimal planner lands a median of +2 laps from the team in Formula 1 (+1 to +3 across horizons and field); a +4 residual in one-stop races shrinks when the planner is asked earlier. In endurance the replay cannot test the tyre model. Run to the flag, the same planner is +10 late. Fourth, 4 defensible rules for the rival agree on 17 of 271 decisions, though the recommendation barely moves between them.

Keywords: race strategy; tyre degradation; out-of-sample validation; motorsport analytics; Formula 1; endurance racing; pit stop optimisation; retrospective audit

1 Introduction

A pit stop is a decision under uncertainty with a measurable cost. The literature on optimising it is small but real, and this paper's contribution is not another optimiser. It is a pair of validation questions that the existing work, for understandable reasons, does not ask.

The first is *transfer*. A degradation model is fitted on races that have happened and used on a race that has not. Whether a slope fitted on past seasons of a circuit predicts the next season of that circuit is an empirical question, and answering it requires holding a season out — which in turn requires more than one season, and ideally more than one championship.

The second is *agreement with practice*. A model that recommends a stop lap can be compared to the lap a team actually chose. This is not a claim about which was faster; it is a measurement of where a model and a professional pit wall part company, and by how much.

Both questions need breadth to answer, and breadth is what this work supplies: one protocol, four championships, 7 car classes, every quantity measured from published timing or absent.

The protocol is the object of this paper and the findings are what it returned. Two of them came back negative, which is a property of the answers rather than of the apparatus. The cost falls in the other direction: my modelling is simpler than much of the work I cite, I ignore energy management entirely, and nothing here makes an existing optimiser better.

2 Related work

Race-strategy research is largely about building the optimiser. The methods vary widely — discrete-event simulation, Monte Carlo, dynamic programming, mixed-integer programming, reinforcement learning, game theory — and the goal is consistent: compute a better plan than a team would otherwise choose.

Bekker and Lotz [2009] give the earliest sustained operations-research treatment, simulating a full race with overtaking and traffic under the refuelling rules of the time. Heilmeier et al. [2018] introduce the lap-time decomposition that much of the later work rests on, combining fuel mass, tyre degradation and driver effects, and Heilmeier et al. [2020a] extend it to a probabilistic simulation sampling accidents, safety cars and lap-time variability. Heilmeier et al. [2020b] then train neural networks on that simulation's output to make calls in real time. Carrasco Heine and Thraves [2022] give an exact dynamic program over stop laps and compound choice, extended to a stochastic version, and Aguad and Thraves [2024] formulate the problem as a zero-sum feedback Stackelberg game between two drivers, finding that a strategic competitor gains over 15% in winning odds. Outside Formula 1 the same machinery has been built for endurance racing, and almost entirely for electric cars, where the binding constraint is charge rather than fuel. van Kampen et al. [2022] solve endurance stint planning and powertrain operation jointly under thermal limits; van Kampen et al. [2024] extend that to model predictive control that prices a competitor's likely response; and de Vries et al. [2026] pose the whole problem as a two-agent game, solving a game-theoretic optimal control problem per lap and training reinforcement learning agents to allocate energy and schedule stops over a race. Boettinger and Klotz [2023] is the rare treatment of GT racing rather than single-seaters, simulating the Nordschleife and learning a stop policy from fuel mass and race position. Recent Formula 1 work adds learning: Thomas et al. [2025] apply explainable reinforcement learning to compound and stop timing,

Fieni et al. [2025] combine a mixed-integer program with an RL agent over energy, tyre wear and pit timing, Fieni et al. [2026] extend that to self-play among competing agents, and Todd et al. [2025] forecast tyre energy from team-internal telemetry. Santillana [2026] deploys a calibrated real-time Monte Carlo engine live, generating natural-language briefings faithful to its own probabilistic state.

Two things are rare across all of it.

The first is out-of-sample validation of the fitted parameters. A tyre model is fitted and its quality reported on the data it was fitted to. Cappello and Hoegh [2026] come closest, building a Bayesian state-space model of degradation from the same public timing source used here, with lap time as fuel mass plus a latent tyre state and pit stops as resets. They also validate out of sample, and are the only work here that does: Cappello and Hoegh [2026] evaluate predictive robustness on 19 race sessions of the 2025 season, by rolling-origin cross-validation against an AR(1) benchmark. What separates that from this paper is the axis along which the held-out data varies. Their held-out sessions are one season, one driver and one championship, held out forward in time; the protocol here holds out a whole season of a circuit-class and asks whether the remaining seasons predict it, over 51 circuit-classes and four championships. Both are out-of-sample. Only one changes season, circuit and class at once. They also report compound-specific degradation differences as not statistically distinct. That is consistent with the instability I measure at scale.

The second is comparison against what teams actually did. Optimisers get compared to other optimisers, to author-constructed baselines, or to the optimum under the model's own assumptions. Bekker and Lotz [2009] compare to real races, and Aguad and Thraves [2024] compare strategic against non-strategic agents inside their own game, but I am not aware of work that replays a large sample of real pit-stop decisions and reports where the optimiser and the pit wall diverge.

A third gap is narrower and sits between the two. A model's assumptions can be tested against measured parameter values without testing the model itself. Salminen [2026] solves the two-car one-stop game analytically and derives the earliest lap on which an undercut cannot be answered, which is the only mechanism I have found that predicts stopping *before* the time optimum — the direction of my third result. His overcut defence requires a tyre advantage of $(o + \varepsilon)/\theta$ laps. At the degradation rates fitted here it is unreachable on a substantial share of circuit-compounds, which I report in the repository because it tests a model that is not mine.

His position-keeping defence is a different matter: it is the third candidate explanation for my own Result 3, so I implement it, test it on the same 357 Formula 1 decisions, and report in Section 7.2 that it fails.

The modelling here is not more sophisticated than Heilmeier et al. [2020a] or Aguad and Thraves [2024]; in places it is deliberately simpler, and it ignores energy management entirely. What is different is the protocol: of the 17 papers catalogued for this review, Cappello and Hoegh [2026] is the only one that validates its fitted parameters on data it did not fit, and none applies a single protocol to more than one championship. I claim no priority. The literature is large and a catalogue is not a census; the claim is only that these 17 papers, each checked against a publication record, do not contain the comparison this paper reports. Two of the four findings that come back from it are negative.

3 Data and protocol

3.1 Sources and scope

Formula 1 lap timing, tyre compound and per-lap track status come from the FastF1 API. WEC, IMSA and ELMS timing comes from a community-maintained endurance dataset carrying lap times, stint structure and flag state but *not* tyre compound. The endurance degradation models are therefore net-slope models where the Formula 1 one is per compound, and that asymmetry is restated wherever the two are compared. Two further Formula 1 sources serve Section 4.2 only: the compound Pirelli nominated for each race, and second-practice laps from the OpenF1 API, which serves the same timing feed from 2023 on. Table 1 gives the scope.

Table 1: Scope. A race-season is one class at one circuit in one season, which is the unit everything here is fitted and reported on.

	Classes	Circuits	Race-seasons	Replayed decisions	Seasons
Formula 1	1	25	104	357	2022–2026
IMSA	3	13	136	632	2021–2026
ELMS	2	9	42	171	2021–2025
WEC	1	11	27	120	2022–2026

Nothing is imputed. Where a source lacks a quantity the affected layer is not fitted, and the gap goes on the record as a gap.

Which laps are modelled. Every fit here runs on a filtered frame rather than on raw timing, and the filters matter enough to state. A lap enters only if it is a green-flag racing lap: in-laps and out-laps are dropped, as are laps under any neutralisation, laps the source marks inaccurate or deleted, and laps whose time is missing. Wet-compound laps are excluded entirely, so every degradation coefficient is a dry-weather quantity. What survives is then trimmed against traffic, by dropping laps slower than a fixed multiple of that driver–race's own median — a relative threshold rather than an absolute one, because the same gap in seconds means different things at Monaco and at Spa — and a stint is modelled only if enough laps survive the trim to fit a shape to. The per-circuit counts at every stage are generated into the degradation reports, so the cost of each filter is visible rather than asserted. What this frame does *not* remove is lap-level variation the filters cannot see: a driver lifting to manage a tyre, an engine mode, a blue flag that cost less than the trim threshold. The driver–race intercept absorbs a constant, not those.

3.2 Degradation

For driver–race i and lap ℓ within stint s, with tyre age a and compound c,

$$y_{i\ell} \;=\; \alpha_i \;+\; \phi\,\ell \;+\; \sum_{k=1}^{d} \beta_{c,k}\, a_{i\ell}^{k} \;+\; \varepsilon_{i\ell}, \tag{1}$$

where y is lap time, α_i is a driver–race fixed effect absorbing car pace, driver pace and track conditions, and $\phi\ell$ is a fuel-burn proxy that is identified only where refuelling is prohibited. In endurance, where cars refuel, ℓ is replaced by laps since the last refuelling stop and the compound index drops out, leaving a single net slope β_1.

Equation 1 is estimated by ordinary least squares through a Moore–Penrose pseudoinverse, which would handle a rank-deficient design — a driver–race seen on one compound only — by a minimum-norm solution. In a rank-deficient design that solution fixes any non-estimable coefficient by the norm rather than by the data, so every reported coefficient was checked: deleting its column lowers the rank of the design in all 116 cases across the Formula 1 circuits, and 0 are non-estimable. The polynomial degree $d \in \{1, 2\}$ is chosen per circuit by leave-one-race-out cross-validation.

Standard errors are cluster-robust (CR1) on the driver–race and referred to $t(G-1)$ rather than the normal [Cameron and Miller, 2015]. The clustering is not decorative: lap-to-lap errors

within one driver's race are strongly dependent, and a coverage simulation on synthetic races with realistic between-unit slope variation puts the classical 95% interval at roughly 75% coverage against 95% for the robust one.

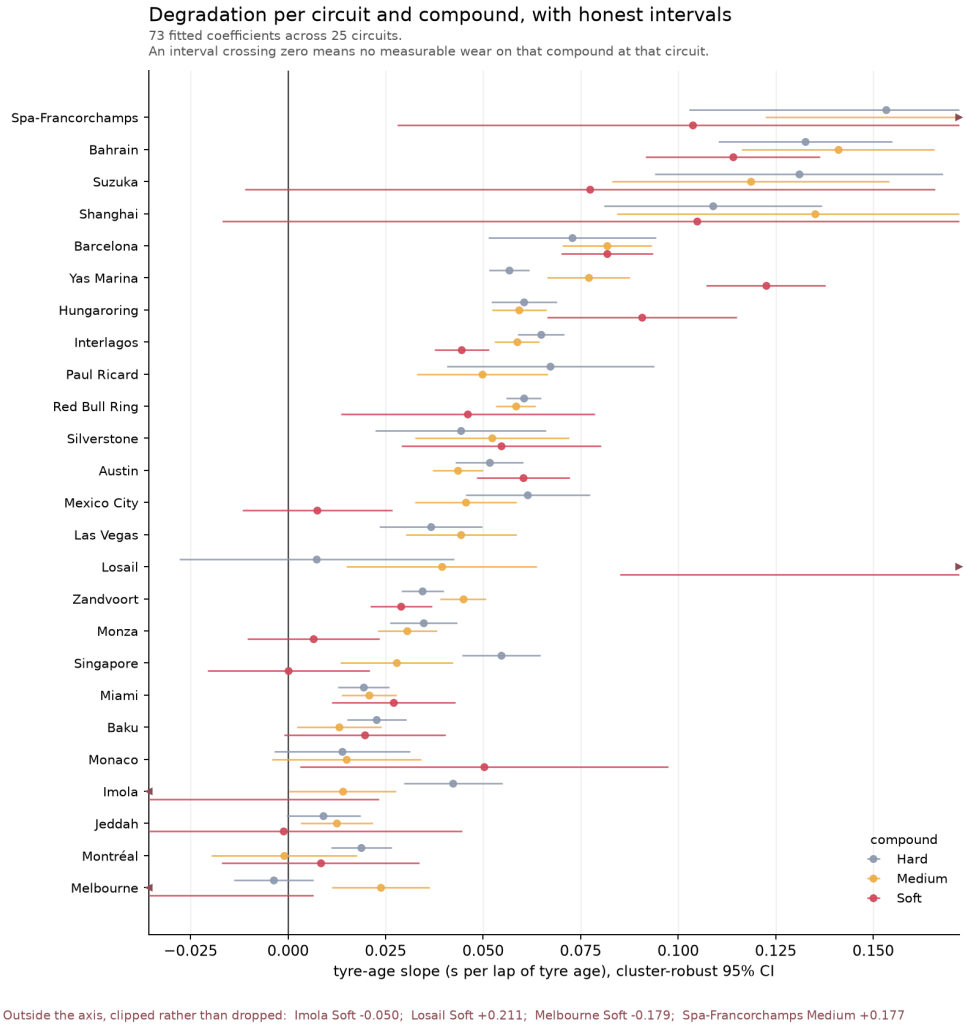


Figure 1: The fitted object itself: linear tyre-degradation coefficients for Formula 1, by circuit and compound, with cluster-robust intervals. 73 coefficients across 25 circuits. Points outside the plotted range are clipped and then named under the axis, because a handful of extremes otherwise compresses every other circuit into a tenth of the width.

Two features of Figure 1 constrain everything after it. 16 of the 73 fitted slopes, 22%, have a cluster-robust interval containing zero: on those circuit–compound pairs I cannot distinguish tyre degradation from none. And within a circuit, 75% of compound pairs have overlapping intervals, so the compound distinction the Formula 1 model carries is largely unresolved by the data behind it. Both are reported, and neither is screened out: a slope that cannot be distinguished from flat is still the measurement I have. Section 9 returns to what this costs.

3.3 Transfer

Fit on every season of a circuit-class but one, predict the held-out season, and score the within-stint demeaned residual:

$$R^2_{\text{within}} \;=\; 1 - \frac{\sum_s \sum_{\ell \in s} \big(\tilde{y}_\ell - \tilde{\hat{y}}_\ell\big)^2}{\sum_s \sum_{\ell \in s} \tilde{y}_\ell^{\,2}}, \qquad \tilde{x}_\ell = x_\ell - \overline{x}_s. \tag{2}$$

The demeaning in Equation 2 is what makes this a test of the *shape* of degradation rather than of pace. A driver–race intercept fitted on training data cannot be observed on a held-out race, so comparing raw lap times would leak information through α_i.

3.4 The multi-stop dynamic program

For an endurance race of L laps with green pace p, net slope β, pit loss π and fuel range F, a plan is a partition of L into stints $(n_1, \ldots, n_{m+1})$ with m stops. Total time under the fitted model is

$$T(n_1, \ldots, n_{m+1}) \;=\; m\pi \;+\; \sum_j \left(n_j p + \beta\, \frac{n_j(n_j+1)}{2} \right), \tag{3}$$

subject to $n_j \le F$ for every stint. The triangular sum is the accumulated tyre loss over a stint of n_j laps. I use it exactly, not as the $\beta n_j^2/2$ integral; they differ by $\beta n_j/2$, which is about a second over a forty-lap stint and is the only quantity the stop decision compares against π.

Because T in Equation 3 is separable across stints given m, the minimum over plans is a one-dimensional dynamic program in remaining laps, solved exactly for each m from the fuel minimum $\lceil L/F \rceil - 1$ upward. A race is called *tyre-limited* when the time-optimal m exceeds that fuel minimum.

3.5 The Monte Carlo engine and the decision audit

The single-next-stop engine evaluates every candidate pit lap over 5,000 shared realisations. Each draw resamples the degradation and fuel coefficients from their cluster-robust posteriors ($t(G-1)$ scaled by the estimated standard errors), the neutralisation hazard from its Gamma posterior, and lap noise at the cross-validated residual scale. Candidates share the same draws, so a comparison between two pit laps carries no unrelated sampling noise. Plans leaving fewer than 3 laps after the stop are excluded. After the stop the engine runs to the end of the race; it plans one stop, not the stops after it, and Section 7.4 measures what that costs the audit.

The audit replays real decisions; it does not forecast them. For every race with a fitted model, the top 5 classified finishers in Formula 1 (5 cars in endurance) who made at least one stop are replayed at their *first* stop. The model is asked 5 laps before that stop in Formula 1 and 3 in endurance, given the state as it was, and its recommended lap is compared to the team's. Stops earlier than lap 8 (6 in endurance) are excluded as incidents rather than strategy. Rivals keep their real, historically observed plans: the decision under study is one car's, the rest of the world is as it happened.

This is what makes the comparison possible and also what stops it being a prediction claim. The decision point is defined relative to a stop that already happened.

3.6 The three baselines

Scored on the same decisions, from the same artifacts, on absolute lap error against the team's stop.

B1, fixed interval. Divide the race into equal stints: $\ell^\star = \mathrm{round}(kL/(m+1))$ for the kth of m stops. Uses no fitted quantity at all.

B2, threshold. Stop on the first lap at which the stint's accumulated tyre loss exceeds the pit loss,

$$\beta \frac{a(a+1)}{2} > \pi, \qquad \text{firing at} \qquad a^\star = \frac{-1+\sqrt{1+8\pi/\beta}}{2}. \tag{4}$$

Equation 4 uses the fitted slope and the measured pit loss, and nothing else — no fuel term, no neutralisation hazard, no distribution over outcomes. The gap between B2 and the full engine is precisely what the Monte Carlo buys.

B3, fuel deadline. Run the tank out. Undefined in Formula 1, where refuelling has been prohibited since 2010, and reported as undefined rather than substituted with the race end.

Where a rule cannot fire — B2 on a tyre that never accumulates enough loss before the flag — no lap is returned and that decision is not scored for that rule. Coverage is reported alongside every error, because a rule answering a fifth of the decisions and landing close on those has not beaten one that answers all of them.

4 Result 1: transfer is a property of the circuit-class

Figure 2 shows the distribution. The maximum is +0.573 at Lime Rock GTD; almost everywhere else a slope fitted on past seasons predicts the next one no better than that season's own mean.

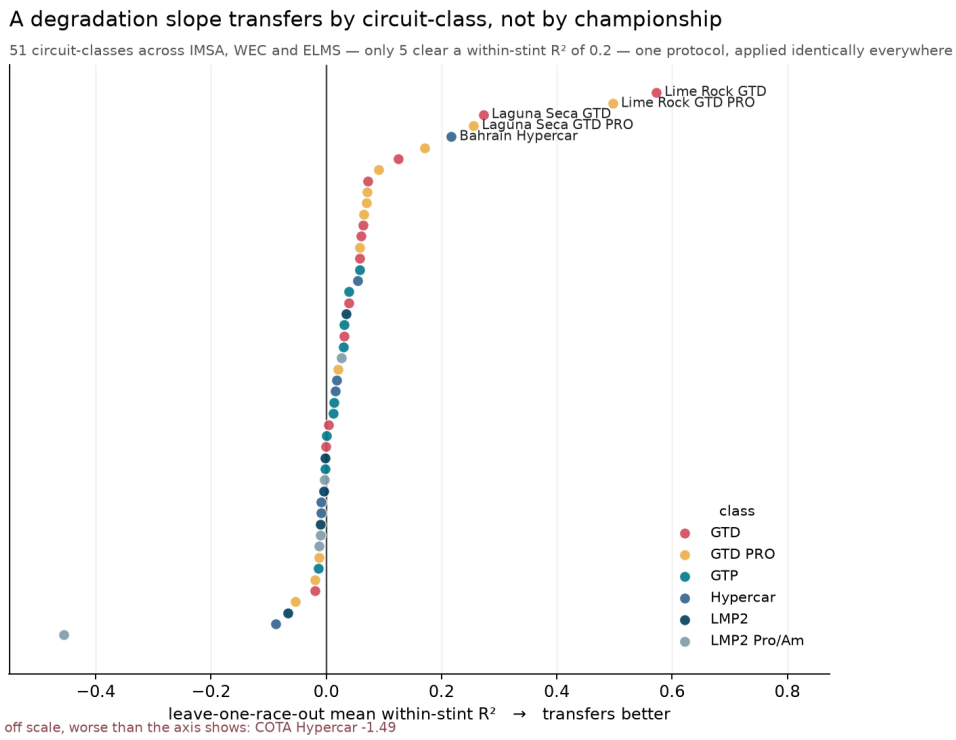


Figure 2: Leave-one-race-out mean within-stint R^2, one point per circuit-class. Of 51 measured, 5 clear 0.2.

Grouping by car type, GT3 classes average +0.104 (95% CI [+0.050, +0.170]) against -0.060 ([-0.199, +0.021]) for prototypes. The difference is +0.164 with a 95% cluster bootstrap interval of [+0.059, +0.312], and a permutation test on the group labels gives $p = 0.0011$. Both are reported because they fail differently: the bootstrap assumes the clusters resemble their population, the permutation assumes only exchangeability under the null. Resampling is at the circuit-class, never the fold: folds inside a circuit-class share a pooled fitted slope, and treating them as independent would roughly quadruple the apparent sample.

The obvious explanation for instability is heterogeneous, performance-balanced machinery. I can rule it out. ELMS LMP2 is a near-specification class — one chassis, one engine, no Balance of Performance — and its slopes fail to transfer exactly as the balanced GT3 fields do. Balance of Performance is therefore not needed for the instability. That is narrower than saying the car plays no part: the GT3–prototype difference above is a car-type effect, and it is confounded with circuit, since the circuits where transfer works do not all host every class. Where transfer does occur it tracks short circuits with cheap stops, which is the same axis Result 2 turns on.

4.1 What a stable slope could have scored

A low score has two explanations that are indistinguishable from outside the measurement. The slope moved between seasons, or there was never enough signal in the laps for any slope to be recovered. Counting how few circuit-classes clear a threshold cannot separate those, and the separation is the result.

So I measured the second one directly, on Formula 1, where the per-compound fit and the lap data are richest. Lap times are rebuilt from each circuit's own fit — its driver–race intercepts, its fuel slope, and *one* degradation slope used for every season, so the slope is stable by construction — plus residuals resampled in whole stints, which keeps their within-stint correlation rather than handing the model independent draws it would find easier to predict. Running the real protocol on that gives the within-stint R^2 a perfectly stable slope earns against the noise actually present.

The ceiling is far lower than the threshold, and it varies by an order of magnitude between circuits: from +0.005 at Silverstone to +0.420 at Jeddah. At 18 of 22 circuits a perfectly stable slope could not have cleared 0.2. The threshold in Figure 2 is therefore reporting how much degradation signal a circuit carries at least as much as it reports stability, and the circuit-classes that clear it are partly the ones where the quantity is measurable at all.

Taking the gap between ceiling and measured score instead — the shortfall the noise does not explain — the picture is sharper than the count it replaces. Of 22 circuits, 8 have slopes that genuinely move, 3 have slopes that are stable once the noise is priced in, 7 carry no recoverable signal either way, and 4 sit between. Instability is real and it is the largest group; it is not, as the raw scores imply, nearly universal.

The degree is chosen with the held-out season in view. The folds above select the polynomial degree by averaging cross-validated RMSE over every fold, including the season then held out, which is a mild leak. Choosing the degree inside each fold instead moves the mean score by 0.0188 and at most 0.2290. The leak is real and small. It is also not free to close: at one circuit, per-fold selection takes the quadratic and extrapolates it off a cliff, which is why the published protocol chooses once. 2 circuits are excluded from the counts above as degenerate fits and named in the generated report.

4.2 Two repairs a team would try first

If slopes move between seasons, two repairs suggest themselves, and both can be tested on Formula 1 with the protocol unchanged.

Code the tyre by what it was. The Formula 1 model carries one slope per compound label, and the labels are relative. Pirelli nominates three compounds of its range for each race, and the same circuit can receive a different three from one season to the next, so last season's Medium may be this season's Hard. I collected the nomination for all 92 races from 2022 to 2025 from the race articles of Wikipedia, which cite Pirelli, and checked it against Pirelli's own press releases, which state it in text for 50 of them and agree every time. When Pirelli added a compound in 2023 it renamed its hardest one; a 2022 lap on that tyre is coded with the name the same rubber carried afterwards, not with its old one. Recoding every lap to the real compound and rerunning the leave-one-race-out protocol, both codings scored on the same laps, changes nothing where the allocation stayed the same. It changed at 16 of 24 circuits; 2 of them are the degenerate fits Section 4.1 excludes, and they are excluded here for the same reason. At the other 14 the median score moves from 0.003 to -0.005, a median paired difference of -0.008 [-0.027, +0.001]; the real compound scores better at 4 circuits and worse at 10, and its largest gain anywhere is +0.017. The test matters most at the 7 of those circuits Section 4.1 calls unstable, where a better coding had something to repair: there the difference runs from -0.349 to +0.017, better at 3 and worse at 4. With every circuit forced to the linear model, so that no quadratic can extrapolate on a compound seen in few seasons, the difference is -0.003 [-0.021, +0.003]. The recoding has a cost the relative labels hide: on average 3.1% of a held-out season's laps, and up to 10.7% at one circuit, are on a compound no other season of that circuit used, and cannot be predicted at all. The claim this supports is narrow. Recoding within a circuit gives each real compound fewer seasons to be fitted on than its relative label had, and it does not repair transfer. A model that pools information across circuits, a circuit severity times a compound effect for instance, is the version that could profit from knowing the real compound, and it is not tested here.

Measure it on Friday. A team need not rely on other seasons at all: in second practice every car runs long runs on a heavy fuel load, on that weekend's tyres and asphalt, though not necessarily on the race load and on a track with less rubber laid down than on Sunday. A long

run here is a practice stint long enough to fit a slope to once its out-lap, in-lap and cool-down laps are dropped, and a compound's Friday slope is used only when several cars ran one on it; on a sprint weekend the only practice session is used, which is the case in 14 of the 56 races. Inside a stint tyre age and fuel burn advance together on Friday as on Sunday, so the Friday slope is the quantity the race model predicts inside a stint and needs no fuel correction. Scored on the same race laps, it does worse than other seasons: a median within-stint R^2 of -0.502 against 0.021, lower in 44 of the 56 races, with a median paired difference of -0.541 [-1.127, -0.236] when whole circuits are resampled. Part of that is a level difference: Friday long runs degrade faster than the same compound does in the race, by a median of +0.059 s a lap, which is the direction a greener track would give, though these data do not separate the causes. Subtracting that gap, learned on every other circuit so that no race informs its own correction, still leaves -0.339. Averaging the corrected Friday shape with the other-seasons shape gives -0.054, worse than other seasons alone in 39 of 56 races [-0.119, -0.022]. A Friday slope rests on a handful of runs per compound by drivers managing fuel, engine modes and traffic in ways the race does not repeat. A team reading its own Friday has information this protocol lacks, but the timing data alone carry less about Sunday's degradation than other seasons do. Neither repair moves Result 1.

4.3 Three baselines a practitioner would ask for

Leave-one-season-out trains on later seasons as well as earlier ones, which no team can do, and a flat slope is a weak reference. On the 22 Formula 1 circuits, with the linear model throughout so that the comparisons are between predictors rather than degrees:

Past seasons only. Predicting each season from the seasons before it, the question a team actually faces, gives a median score of -0.019 against 0.009 for leave-one-season-out, a median paired difference of -0.035 [-0.114, +0.005], worse at 15 circuits and better at 7. Fewer training seasons and no later ones cost a little; they do not change the picture.

The circuit against the category. A slope common to the category — for each compound, the median over every *other* circuit of that circuit's linear fit, with the median fuel slope — scores 0.004 against 0.009, a median paired difference of -0.012 [-0.087, +0.075], better at 10 circuits and worse at 12. A circuit's own history does not predict its next season better than a slope that knows nothing about the circuit.

The ceiling as the yardstick. Measured against what a perfectly stable slope could have scored at the same circuit, rather than against a fixed threshold, the median score at the 15 circuits where the ceiling exceeds 0.05 is 0.71 of it, and 9 reach half. That is the more informative reading of Result 1 for Formula 1: where the laps carry signal at all, a slope from other seasons recovers much of what a stable one could, and where they do not, no count above a threshold says anything about stability. The ceiling has not been built for the endurance classes, so the endurance count stands on the raw threshold alone.

Two further checks concern the measurement rather than the predictor. Within a stint, tyre age and lap number advance together, so the within-stint R^2 scores the net slope — degradation plus fuel burn plus whatever the track does during the stint — and Result 1 is a statement about that net quantity. And the traffic trim, which drops laps slower than a multiple of the driver's race median, truncates the response itself; refitted with the multiple at 1.05 and 1.20 instead of 1.10, at most 17% of slopes move by more than a tenth and the median score ranges from 0.007 to 0.033.

5 Result 2: the cost of the stop sets the strategy regime

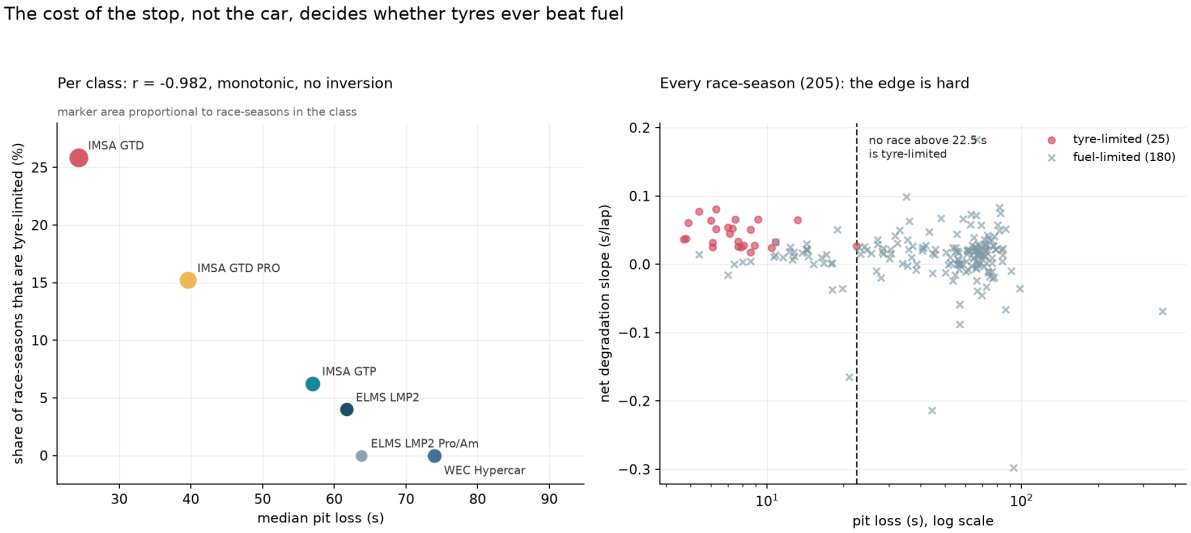


Figure 3: Left: class-level median pit loss against the share of race-seasons where the tyre-optimal plan needs more stops than the fuel minimum. Right: every race-season, with the empirical edge marked.

Figure 3 plots the rule. An exact dynamic program plans each endurance race under a hard fuel constraint and reports whether the tyre-optimal plan requires more stops than the fuel minimum. Across 205 race-seasons, 25 are tyre-limited.

The class-level correlation between median pit loss and tyre-limited share is -0.982 with a 95% interval of [-0.986, -0.745], monotonic across all classes with no inversion. That interval is bootstrapped over *races*, recomputing each class summary inside every replicate; bootstrapping the class summaries themselves would give an interval over six fixed championships, which is not a source of variation that exists.

At race level, 150 of 205 race-seasons have a pit loss above 22.5 s and 0 of them is tyre-limited, across every class.

Is this circular? The optimiser decides the regime by comparing what an extra stop costs against what the tyre gives back, so pit loss predicting the regime is pit loss predicting one side of its own criterion. The objection is fair and the answer has to be a measurement. If the stop cost were doing all the work, the degradation slope would carry no information about the regime once the stop cost is known. It carries some. Ranking the 205 race-seasons by pit loss alone orders the 25 tyre-limited against the 180 fuel-limited ones at an AUC of 0.977 (95% CI [0.955, 0.994]); by net degradation slope alone, 0.865 ([0.806, 0.915]). Both are well clear of a coin toss and the ordering between them is not close.

This does not establish that pit loss *causes* the regime, and I do not claim it: the two are related by construction. What it establishes is that the comparison is asymmetric in practice. Knowing the cost of a stop tells you nearly the whole answer across four championships; knowing how fast the tyre wears tells you considerably less. The case that pit loss is exogenous to the race outcome is made separately, from the fact that it varies far more between circuits than between seasons of the same circuit.

Is the separation a class label, or the spread of one predictor? Two objections sit here and both have measurements rather than arguments.

The first is that pit loss simply varies more than the slope does in this sample, so it wins a ranking contest for that reason. A rank statistic is invariant to rescaling either variable (standardising pit loss leaves it at 0.977), but that answers only the units version of the objection; it says nothing about how widely each varies against the boundary. The closed form makes the question precise. Up to the dynamic program's details, an extra stop pays when $\beta n^2/2$ exceeds π, so the regime is set by the index $\log\beta + 2\log n - \log\pi$, with n the fuel range in laps. On the 164 race-seasons with a positive net slope, which hold every tyre-limited one, that index separates the regimes at 0.996. Its variance splits into 51% from the slope, 39% from pit loss

and 10% from the stint length: in spread, the slope term is the larger. But much of that spread lies far from the boundary, among slopes close to zero whose logarithm runs off without changing any verdict. Dropping each term from the index in turn measures what moves races across it: without pit loss the index falls to 0.923, without the slope to 0.953. Both terms enter; in these four championships the stop cost decides more of the verdicts. That is a description of this sample, not a law of motorsport, and a sample in which pit loss varied less would give the slope more of the answer.

The second is sharper. Pit loss is nearly constant inside a class, so its AUC could be measuring class identity wearing a continuous disguise — and the class-level correlation above rests on one point per class. Ranking race-seasons by their class's mean pit loss alone gives 0.760, well short of 0.977 for pit loss itself, so the label does not carry the result. Nor is pit loss mostly a class property: 92% of its variance is *within* class rather than between, which is what one would expect of a quantity set by the geometry of a pit lane rather than by a car. Recomputed inside each class, where class identity is held fixed by construction, the cost of the stop still separates the two regimes in all 4 classes where both regimes occur, between 0.963 and 1.000. The remaining 2 classes contain one regime only, so no AUC is defined for them and none is reported.

The third objection is precision. Pit loss is measured from stop durations; the slope is estimated, and a noisily measured predictor loses AUC even when the quantity behind it matters as much. If that were the cause, the slope's AUC should rise as the least precise slopes are dropped. It does not. Keeping the 103 race-seasons with the narrowest slope intervals gives 0.872 for the slope against 0.964 for pit loss, and keeping the most precise 68 gives 0.815 against 0.964, though with only 8 tyre-limited race-seasons left in that subset. The gap is not an artefact of how well the slope is measured.

What the median pit loss is a median of. It is taken over every green-flag stop in the race-season, which in endurance racing includes stops that change driver as well as stops that do not. A driver change is slower in every class measured, so the class medians are not a measure of tyre-service time alone. The class *ordering* is unchanged whether the median is taken over all stops, only those without a driver change, or only those with one, which is what the correlation above depends on.

The condition behind it. A linear slope β over a stint of n laps accumulates $\beta n(n+1)/2$ seconds of degradation, so an extra stop repays its π seconds when that sum exceeds π. I state it because it is the readable form of the mechanism, and I do not use it: the dynamic program above is exact under the fuel cap and the stop count is read from its solution. The closed form is an approximation to that, checked against it rather than substituted for it.

The rule is robust; the constant is not. The edge sits at 22.5 s because a single race puts it there; the next tyre-limited race is at 13.2 s, a drop of 41%. I do not report a bootstrap interval for it, because the statistic is a maximum: it sits on the boundary of its own support, so the bootstrap distribution is degenerate above it and understates uncertainty below. Quote the rule; treat the number as an order of magnitude.

6 Inference

Each interval above is a percentile bootstrap resampled at the level its data varies at, and those levels differ: circuit-classes for the transfer comparison, races for the pit-loss correlation. The choice is not cosmetic. Resampling transfer folds instead of circuit-classes would treat 51 clusters as roughly four times that many independent observations, and the interval would narrow for no reason connected to the data. Figure 4 collects every result that carries one.

The audit's central finding is a median over decisions that are not independent: several drivers in one race share a track, a Safety Car and one tyre allocation, so a bootstrap over decisions would understate its width in the direction that flatters the result. The audit tables carry race identifiers, so the race is the unit resampled.

For the first-run audit of Result 3 (Section 7), 2 of the 7 class-level medians contain zero (HYPERCAR, LMP2). Formula 1 is +10 laps with a race-clustered interval of [+8.0, +11.5] over 74 races; IMSA GTD +12 [+8.0, +14.0]; IMSA GTD PRO +13 [+9.0, +16.0]; IMSA GTP +9 [+5.0, +15.0]; WEC Hypercar +1 [+0.0, +1.0]; ELMS LMP2 +2 [-1.0, +3.0]; ELMS LMP2 Pro/Am +3 [+1.0, +4.0]. As first run, the audit therefore resolved a disagreement in Formula 1 and the three IMSA classes and none in WEC Hypercar and ELMS; Section 7.4 shows that the resolved part is mostly a property of the audit.

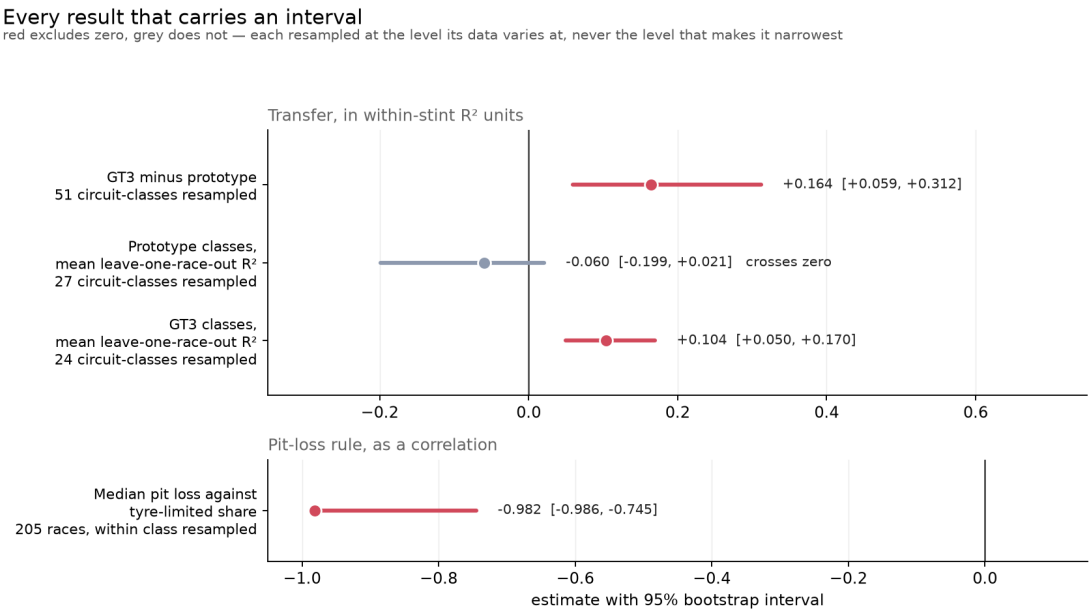


Figure 4: Every result carrying a bootstrap interval. The two panels are separate because a within-stint R^2 difference and a correlation are not measured on the same scale, and a forest plot works by letting the reader compare bar lengths. The cheap-stop edge is absent by design: it is a maximum, and a leave-one-out sensitivity is reported for it instead of an interval it cannot support.

7 Result 3: most of the planner–team gap was the question the audit asked

The planner replayed here is the single-next-stop engine of Section 3.5. It chooses the lap of the next stop to minimise one car's expected race time against a linear degradation slope, a measured stop cost and a neutralisation hazard, and it runs the stint after that stop to the flag. It represents neither the tyre allocation, nor traffic, nor energy management, nor a championship position worth protecting. Every sentence below should be read as a disagreement between a model and a practice rather than as evidence about which was faster.

The result first. With each team's own next stop held fixed, the planner lands a median of +2 laps from the real stop in Formula 1 [+1.0, +4.0], and the interval contains zero in every endurance class. What survives is +4 laps [+2.0, +7.0] in one-stop races, where the planner and the pit wall were always answering the same question (Section 7.4).

The route there is a methodological finding of its own, and the rest of this section follows it. Run as a single-next-stop audit, which lets the planner stretch its second stint to the flag, the same planner is +10 laps late in Formula 1. Four explanations for that gap were tested and failed before the fifth showed that the audit had been asking a different question from the one

the team answered. A comparison between an optimiser and practice has to hold the team's committed strategy fixed, or it measures the difference between the two questions.

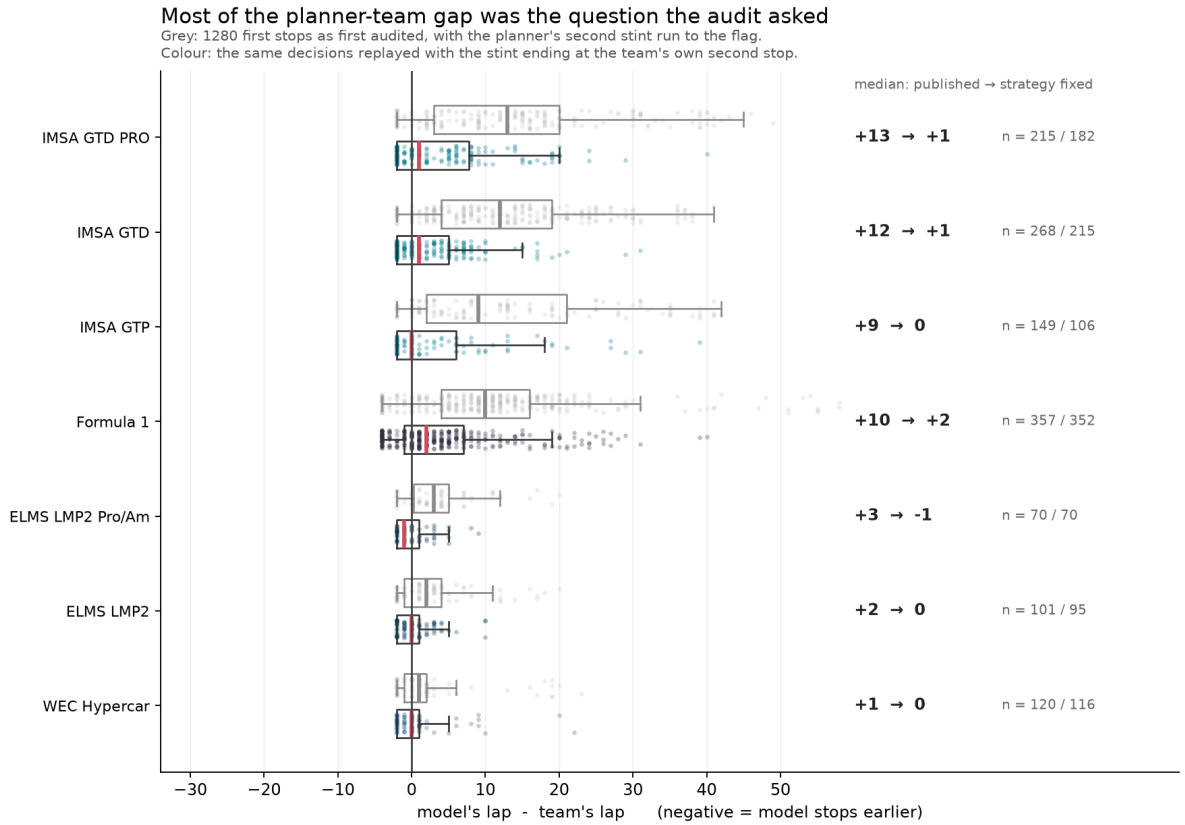


Figure 5: Model lap minus team lap, 1,280 replayed first stops. Grey: the audit as first run, with the planner's second stint run to the flag. Colour: the same decisions replayed with the stint ending at the team's own second stop (Section 7.4). In Formula 1 a one-stop decision keeps its first-run value; in endurance only decisions with a later stop can be replayed, hence the smaller counts. Boxes show quartiles; the cloud behind each is the sample.

Figure 5 shows the whole distribution. Every real first stop with a fitted model is replayed. The model is asked five laps before the stop, given the state as it was, and its recommended lap compared to the team's. This is a replay, not a forecast: the decision point is defined relative to a stop that already happened, which is what makes the comparison possible and also what stops it being a prediction claim.

In Formula 1 the optimiser stops later on 80% of decisions, by a median of 12 laps on those; over every decision, late or not, the median is +10 laps. The endurance figures below are medians over every decision, so +10 is the comparable one. In endurance the gap tracks the caution rate: WEC +1 laps at 8% of first stops taken under caution, ELMS +2 at 20%, IMSA +12 at 53%.

Read that ladder with the race-clustered intervals in Section 6 beside it. Its two lowest rungs, WEC Hypercar and ELMS LMP2, have intervals containing zero: the ordering is real

but the small gaps at the bottom of it are not established, and only the Formula 1 and IMSA disagreements are — as first measured, before Section 7.4 removes most of them.

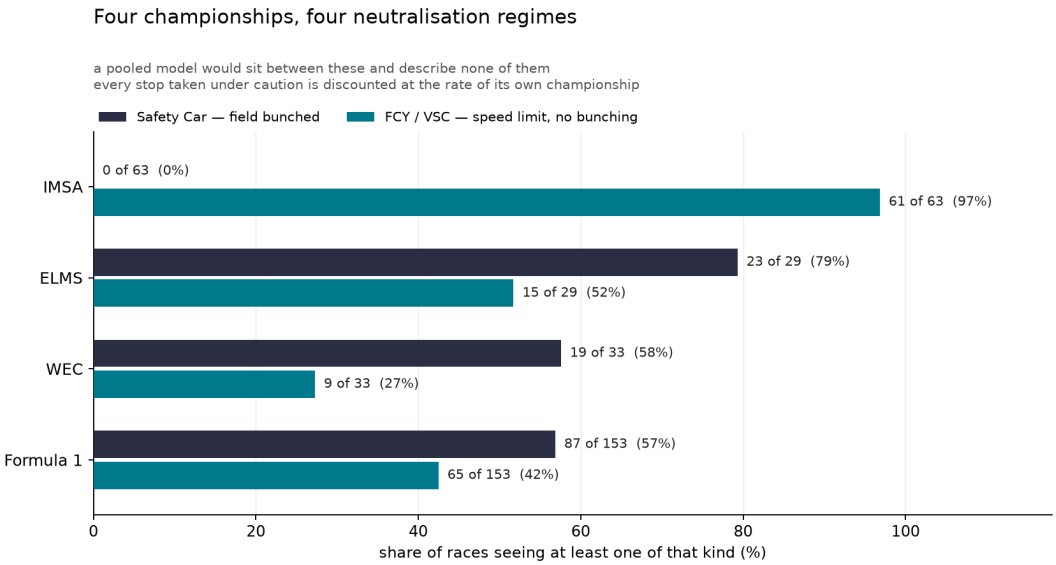


Figure 6: How often each championship neutralises a race, split by kind. A Safety Car bunches the field; a Full Course Yellow — a Virtual Safety Car in Formula 1 — imposes a field-wide speed limit without bunching it, and the two discount the cost of a stop by very different amounts. IMSA deployed no Safety Car in any scoped race and neutralises almost every race by the other route, so a figure showing only the first column would put IMSA at zero and contradict its caution share above. The unit is the race, not the car: the flag is the modal per-lap flag across the field, never one car's timing row. Classes are not separated because every class in a race shares the same neutralisation.

Figure 6 gives the neutralisation rates behind that ladder. A stop taken under caution costs a fraction of a green-flag stop, so a championship that neutralises often gives teams a reason to stop early that the optimiser, which prices the stop at its green-flag cost, does not see. That ordering is consistent with the gap, and it is not a test of it: three championships is three points, and I do not fit a line through them.

7.1 Testing the two candidate explanations

I proposed two mechanisms and then attacked them.

Absence of track position. The engine optimises one car's expected race time and cannot pay for an undercut, which is exactly why a team stops before it has to. Re-running all 357 Formula 1 decisions through a cover-aware Stackelberg engine consuming each circuit's measured overtaking difficulty moves the recommendation *away* from the real stop, not toward it. **Rejected.**

Across the calendar the cover-aware recommendation lands a median of -1 lap further from the real stop than the single-car one, and only 18% of decisions improve at all.

Two separate things follow and they should not be read together. The first is about the hypothesis: modelling the cover does not move the recommendation toward what teams did, so an unmodelled undercut does not account for the gap. That holds under every rival-selection rule in Section 8, including the one that fits the engine best. The second is about the test rather than the hypothesis: part of the *further* in that sentence was an artefact of how the rival was chosen. Under the published selection the cover-aware model is a median -1 lap worse than the single-car one, which reads as an engine that damages its own input; under the standalone-time selection it is level at +0 and improves 99 decisions against 63. The hypothesis dies either way. It dies cleanly under the better selection, and the earlier appearance of a model actively making things worse was in part a measurement defect I have since fixed.

Two qualifications stand between the first of those and a clean refutation.

The first is that the covariate is thinner than its headline suggests. Figure 7 shows a 14-fold range across 25 circuits, which sounds decisive and is carried by its two ends. Monaco sits at 0.0047 swaps per lap, a factor of 4.1 below the next circuit, with a spread across its 4 races of only 0.0016 — the one entry that is genuinely separated. The top of the range rests on 3 races at a circuit first run in 2023 and is the noisiest row in the table. In between, the standard deviation across circuits (0.0104) is about the same as the median standard deviation across races within a circuit (0.0103), so the twenty circuits in the middle are not separable from one another on this measure.

The second is that the mechanism is not uniformly wrong. At Monaco, where overtaking is hardest and the covariate resolves cleanly, the cover-aware engine does move toward the real stop, by a median of +0.5 laps over 8 decisions, and the rank correlation between swap rate and improvement is -0.16 across the calendar. Both point the way the mechanism predicts. Neither is remotely large enough: half a lap against a median discrepancy of +10 laps is the right sign at the wrong order of magnitude. Track position is doing something. It is not doing this.

Slopes biased toward durability. A tyre that looks flatter than it is makes staying out look cheaper. Measured against an independent source that separates tyre wear from fuel burn by a different method on different data, the median paired difference is negligible — an error of that size moves a stop by several race distances, not by ten laps. **Not detected.** I note that comparing the two *medians* instead of the median of paired differences produces a small

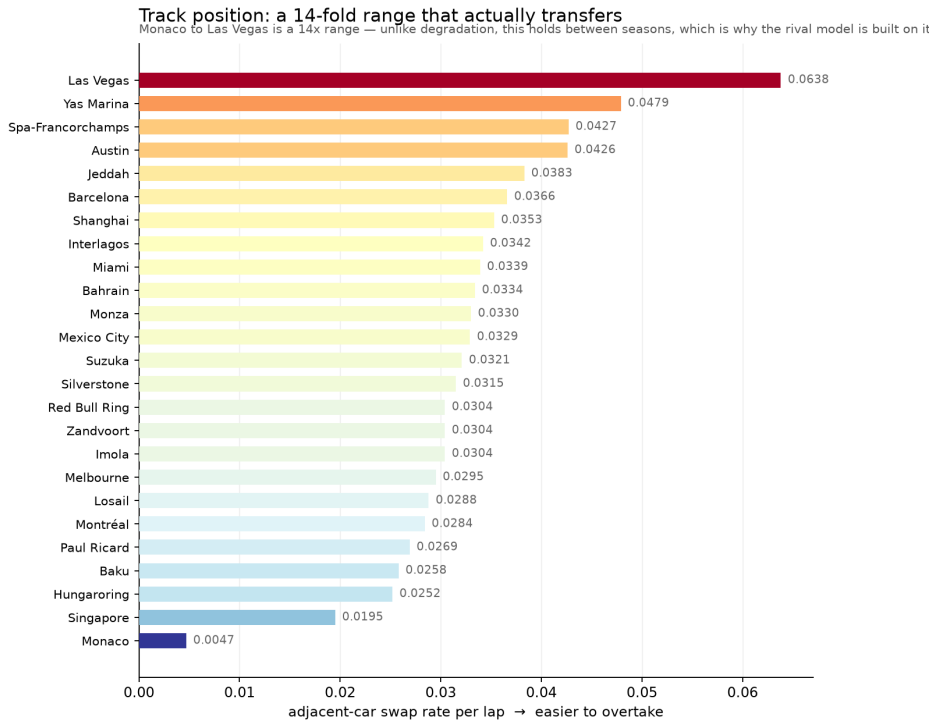


Figure 7: Adjacent-car swap rate per lap, by circuit. This is the primitive the cover-aware engine consumes, shown so that the rejection above can be read against the spread of the input, and not assumed to come from a flat one.

apparent bias that is an artefact of comparing summaries rather than pairs.

7.2 A third explanation, from published work

The two explanations tested above were mine. A third comes from Salminen [2026], who solves the two-car one-stop problem as a perfect-information sequential game and derives the earliest lap on which an undercut cannot be answered. It is the only mechanism I have found that predicts stopping *before* the time-optimal lap, which is the direction of the disagreement reported here: a chaser stops when the leader's answer expires, and waiting for the time optimum hands the leader a window.

The leader has two answers in that model. The overcut needs a tyre advantage of $(o+\varepsilon)/\theta$ laps, where o is an overtaking threshold in seconds. I hold no such quantity — what this project measures per circuit is an adjacent-pair swap probability per lap, a different object that does not convert into one — so I implement only the position-keeping defence, which is efficient while $\theta l_u + \varepsilon < g$ and therefore expires at a tyre age of $(g-\varepsilon)/\theta$. Where the overcut is unavailable, the earliest undefendable undercut is the lap after that. I set $\varepsilon = 0$ throughout: estimating it would mean reading a chaser's pace while it sits in dirty air, which inflates it, and a larger ε

expires the defence earlier and so moves the prediction toward the real stop.

It fails, and by how much. Across the 254 decisions carrying a prediction, the earliest undefendable undercut lands a median of +55 laps from the real stop, against +9 for the single-car optimiser on the same decisions. It is closer on 41 and further on 207. Fed with the gaps and slopes measured here, the condition says a Formula 1 leader can defend an undercut for longer than the race lasts, so a chaser should never undercut at all.

The failure has a number rather than a story behind it. Setting l_u to the tyre age of the stop a team actually made gives the degradation rate at which the defence would have expired exactly then, $\theta^* = (g - \varepsilon)/l_u$. Its median is 0.185 s/lap against a measured 0.045, a ratio of 3.4 with quartiles 1.4 to 8.6, and θ^* exceeds the measured rate on 79% of decisions. A rate of 0.185 s/lap sits at the 99th percentile of every slope fitted in this work. The gap is therefore not a matter of calibration: for the equilibrium to reproduce what teams do, tyres would have to degrade at a rate this data almost never shows.

A reading I have not established. θl_u prices the defender's loss over a single lap on a tyre l_u laps old. At a measured 0.045 s/lap and a tyre the age teams stop at, that is under a second, which the median 3.4 s lead covers easily. What wins a real undercut is the advantage of a tyre on its first flying lap over one at the end of its life, and a linear slope has no term for that. Measuring it directly is a different estimator and I have not attempted it.

What holds whatever the verdict. Of the 357 decisions, 91 belong to a car with nobody ahead of it on the clock within one lap. The race leader has no undercut to perform, so roughly a quarter of the audited set carries no prediction under any two-car model, which bounds what this family of explanations could ever account for. For the remaining 266, the gap to the leader is a median 3.4 s, and on 260 of 266 it is smaller than the circuit's measured pit loss. Whether the position is reachable by stopping first is therefore not the binding constraint, which is the opposite of what I expected to find.

Defects found while building this. All three inflated the defence's window, so all three made this negative result look stronger than it is, and I report them for that reason. Comparing cumulative race time with no bound found rivals a full lap away; the nearest car ahead is now capped at one green lap and the widest retained gap is 36 s. The expiry is a tyre age and was

being compared against a lap number, which agree on only 78% of these decisions. And the breadth layer's pit-loss table names circuits differently from the lap files, which still drops 85 of 266 values if joined on the slug. Each now has a test.

7.3 A fourth explanation, and why it cannot move the lap

The most natural candidate explanation is structural: a linear slope has no one-off gain at a stop, so pricing a jump in pace at the stop should move the recommended lap. It cannot, and neither can its opposite. With the number of stops fixed — one before the flag in the first run, one before the team's next stop in the committed replay — any time gained or lost once per stop is the same for every candidate lap and cancels from the comparison. A one-off term can change how many stops are worth making, not when to make the one being planned.

The data do show such a term, and it has the other sign from the objection. Pooled over 24 Formula 1 circuits, the fit's residual at the youngest measured tyre age is +0.187 s and at the next +0.059 s, where negative means faster than modelled: laps on a new set come in *slower* than the fit expects, and the sign turns over once the set is several laps old. Per circuit, fresh laps are faster than modelled at 7 of 24, median -0.048 s. That is tyre warm-up. Its only route to the timing is through estimation: slow early laps flatten the fitted slope, and because hard compounds warm up more slowly than soft ones the effect on the ratio of two slopes, which is what sets the timing, has no determined sign. It would have to be measured compound by compound, and I have not done so. What this section establishes is narrower than an earlier version of it claimed: the one-off-gain objection fails on structure, and the warm-up it points to does not settle the direction of the gap.

7.4 A fifth explanation: the question the audit asked

The four explanations above all look for something the planner misprices. The fifth is in the audit. The engine plans one stop and runs the stint after it to the flag, so for a car that stopped once it was answering the question the pit wall answered. For a car that stopped again, it was not: the team's second stint ended at its second stop, and an engine made to stretch one set of tyres to the end will place the first stop later for that reason alone.

The published gap splits accordingly. Over the 150 Formula 1 decisions by drivers who stopped once, the median is +4 laps [+2.0, +7.0]; over the 202 by drivers who stopped again it is +12 [+11.0, +14.0], and +17 for three or more stops. Inside the 19 races containing both

kinds of decision the multi-stop median exceeds the one-stop median by +5 laps, so the split is not only a difference between circuits.

I then replayed every decision whose driver stopped again with the race ending, for the engine, at that driver's real second stop. Everything after that stop runs on fresh tyres whichever first-stop lap is chosen, so it is common to every candidate and drops out: the optimum over the shortened horizon is exactly the optimum given the team's own second stop. Nothing else changes; replayed with the full horizon, the same code reproduces every published recommendation in a random sample of forty. Leaving out the 5 decisions whose real stop falls outside the shortened window, and with the team's strategy held fixed, the Formula 1 median falls from +9.5 to +2 laps [+1.0, +4.0], and among multi-stop decisions from +12 to +1 [+1.0, +2.0]. In endurance the same replay, with each stint also bounded by the fuel range, covers 784 decisions, and the interval contains zero in all 6 classes. The first-run medians below are over those replayable decisions only, so they differ slightly from the whole-class medians in Figure 5: IMSA GTD falls from +11 to +1 [+0.0, +2.0], IMSA GTD PRO from +13 to +1 [+0.0, +5.0], and IMSA GTP from +8 to 0 [-2.0, +2.0].

Two objections have to be met before that is read as agreement. The first is hindsight: a second stop taken under a Safety Car was not planned at the decision lap. Keeping only the 155 decisions whose second stop was under green leaves the Formula 1 median at +2 [+1.0, +2.0]. The second is that fixing the second stop shrinks the set of laps the engine can pick from, and a narrow enough window makes agreement automatic. In Formula 1 it does not: the window is a median of 24 laps, inside which a lap drawn uniformly would miss the real stop by a median of 8.3 and the window's midpoint by 7.5, against 3 for the engine. In endurance the engine beats both chance baselines in only 3 of 6 classes, the three IMSA ones (GTD: window 12, engine 2, uniform 4.5, midpoint 4.0). In WEC Hypercar and both ELMS classes the fuel range leaves a window of 4 to 7 laps, the engine does no better than chance, and the replay is uninformative there — which is consistent with those being the classes where the published gap was never established.

What survives is the one-stop residual. Where the engine and the team were answering the same question, the engine still stops +4 laps later [+2.0, +7.0]. That is the part of Result 3 the audit design does not explain.

One caution applies to every committed number. The team's second stop was chosen knowing where the first had been and what happened between them, so conditioning on it is conditioning

on a later decision. The committed median is agreement given the plan the team revealed, which is what makes the planner and the team answer the same question; it is not by itself a measure of how good the planner is.

Do the earlier rejections hold on it? The cover-aware engine and the earliest undefendable undercut were rejected on the first-run audit, whose gap was mostly the audit's question. A mechanism too weak for +10 laps could still account for +4, so both are retested on the 150 one-stop first stops alone, where the committed replay and the first run coincide. Neither recovers. Under the published rival rule the cover-aware engine moves closer to the real stop on 34 of them and further on 74; across all 4 rival rules it moves closer on at most 37 and further on at least 55, a sign test giving $p \leq 0.006$ under each. The earliest undefendable undercut has no prediction for the 39 race leaders; on the 109 decisions that carry one it lands a median of +73 laps from the real stop against +5 for the single-car optimiser, closer on 9 and further on 97. The argument of Section 7.3 does not depend on the size of the gap and needs no retest.

Does the audit's design make the gap? Three choices could. The planner is asked five laps before the real stop, so it can be at most four laps early and arbitrarily late: 13% of committed decisions sit on that bound. Only the top five finishers are replayed, the cars whose strategy worked and that most often ran in clean air. And the median of whole-lap gaps moves in whole laps. So the replay was rerun asked ten and fifteen laps ahead, and on every car running at the flag, with the Hodges–Lehmann estimate beside the median. For the top five the median is +2 [+1.0, +4.0] asked five laps ahead, +2 [+1.0, +4.0] at ten and +1 [+1.0, +3.0] at fifteen, with 7% then on the bound; for the whole field (1,152 decisions asked five laps ahead) it is +3 [+2.0, +3.0], +2 and +1. The overall agreement does not depend on the design. The one-stop residual does: +4 [+2.0, +6.5] asked five laps ahead, +2 [+0.0, +5.0] at fifteen, and for the whole field +4 [+2.0, +5.0] falling to +1 [-1.0, +3.0]. Part of it is the asymmetric window the audit gives the planner.

What the timing depends on. With one stop before a fixed endpoint and linear slopes, the time-optimal stop lap k after the decision solves $\beta_{\text{old}}(a_0 + k) = \beta_{\text{new}}(W - k)$, for a tyre aged a_0 and a window of W laps. It depends on the two slopes only through their ratio; the fuel slope and any cost paid once per stop cancel. This closed form, which leaves out rivals and neutralisations, lands a median of 0.9 laps from the engine's own lap on 243 decisions at linear

circuits, so it describes what the engine is doing. Two consequences follow. The gap could be noise in the ratio, a quotient of two imprecise slopes; it is not, since decisions whose ratio is resolved (its 95% interval excludes one) have a median gap of +2 over 73, against +3 over 187 unresolved ones, and +5 against +3.5 in one-stop races. And the team's real stop implies a ratio: inverting the closed form at the lap the team chose gives a median of 0.77, against 1.02 fitted, and the implied ratio is the smaller in 70% of 232 decisions between two different compounds. Teams stop as if the tyre they are on were wearing faster, relative to the one they are about to fit, than the fit says. That is the one-stop residual restated as a parameter, and it is where a structural estimate in the manner of Romer [2006] would start.

In endurance the same algebra removes the test. The endurance model has one net slope per class, so $\beta_{\text{old}} = \beta_{\text{new}}$ and the deterministic part of the timing reduces to equal tyre ages either side of the stop, whatever the slope is. The endurance replay therefore tests the fuel and neutralisation logic, not the tyre model, and it is no surprise that B1, which divides the race into equal stints, matches the optimiser there.

7.5 What the remaining gap costs

A gap in laps is not a gap in time, and the planner's own price for it is a model output. The committed replay gives both: for each first stop, the planner's lap with the team's next stop fixed, and the median time the planner predicts the team's lap cost against its own. Over the same stretch of the real race, from the decision lap to the team's next stop or to the flag, I measured each car's elapsed time against what its own median green-flag lap predicts for that many laps, and compared cars within a race, which share the track, the weather and the tyre allocation. A race fixed effect absorbs what is common to the field, the window's length is a control, and a window containing a Safety Car, a Virtual Safety Car or a red flag for that car is dropped, since a stop under neutralisation is cheaper by an amount the planner's cost does not describe.

That leaves 213 first stops in 44 races. They sit a median of 3 laps from the planner's lap, a distance the planner prices at a median of 0.61 s. In the races, each lap of distance from the planner's lap goes with 0.42 s of extra elapsed time [-0.24, +1.11], against 0.25 s by the planner's own pricing. Regressed on the planner's predicted cost directly, the slope is 0.40 [-0.15, +2.65], where one would mean the model's costs show up in the clock unchanged. Both intervals contain zero, and both contain the planner's price; the races are consistent with the planner's tenths of a

second a lap and do not rule out a full second. They cannot tell these apart because between cars of the same race the excess time has a median standard deviation of 5.6 s, an order of magnitude more than the planner says the typical gap costs. In power terms, the smallest per-lap cost these 213 decisions could detect at the 5% level with 80% power is about 0.96 s, and detecting the planner's own price would take about 3,100 decisions: the absence of a verdict is a quantifiable shortage of data, not only noise.

This is association, not an experiment. Teams leave the planner's lap for reasons it cannot see — traffic, a rival's stop, a car problem — and those reasons have costs of their own, in either direction. What it establishes is a limit on what the races can say: the residual Result 3 leaves is priced by the planner at tenths of a second per lap of distance, and a race is too noisy to confirm that price or to reject one several times larger.

7.6 Comparison against three baselines

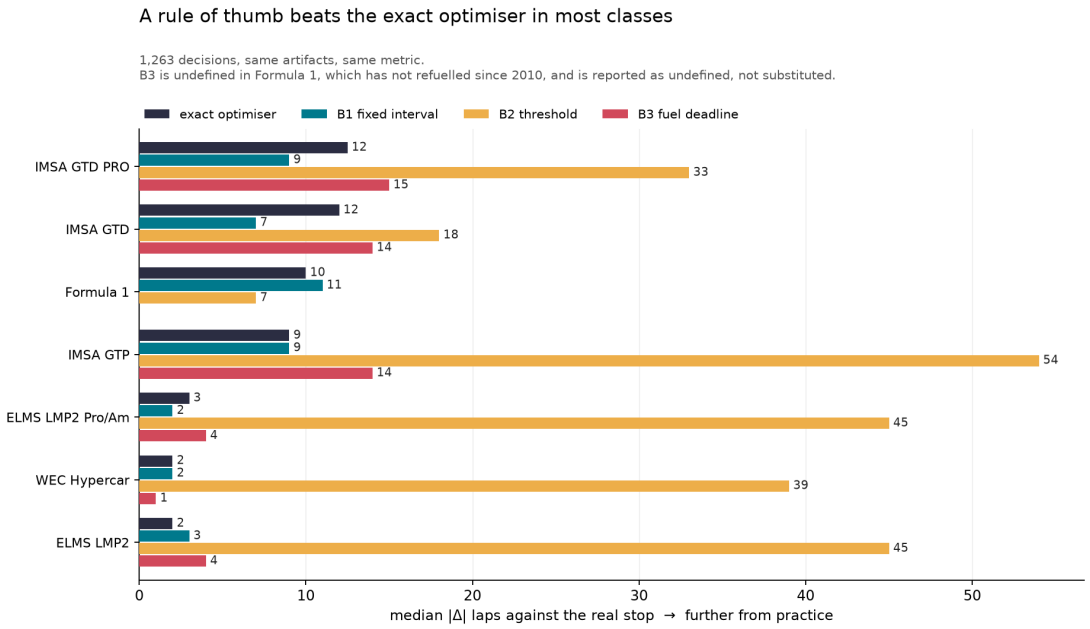


Figure 8: Median absolute lap error against the real stop, one row per class rather than per championship. IMSA's three classes run the same rounds under pit losses of 57, 24 and 40 seconds, so B3 — run the tank out — means a different thing in each and one IMSA row would describe none of them.

Three rules were scored on 1,263 of the same decisions, from the same artifacts, on the same metric (Figure 8 and Table 2). B1 divides the race into equal stints and uses no fitted quantity at all. B2 stops once the tyre's accumulated loss exceeds the pit loss, using the fitted slope and the measured pit loss and nothing else. B3 runs the fuel load out, and is undefined in

Formula 1, where refuelling has been prohibited since 2010 — I report it as undefined rather than substituting the race end.

Results are reported per class rather than per championship. IMSA runs GTP, GTD and GTD PRO at the same rounds with median pit losses of 57, 24 and 40 seconds, so a single IMSA row averages three strategy regimes into a number describing none of them. That grouping also concealed a difference worth seeing: GTP, the IMSA class with the dearest stops, has the smallest disagreement of the three.

Table 2: Median absolute lap error against the real stop, per class. Lower is closer to what teams did. Bold marks the best of the first four columns. The last column is the optimiser with each team's next stop held fixed (Section 7.4), over the decisions that can be replayed; it is given the team's second stop, which no rule is. B3 is undefined in Formula 1, which has not refuelled since 2010.

	Optimiser	B1 interval	B2 threshold	B3 fuel	Strategy fixed
Formula 1	10	11	**7**	–	4
IMSA GTP	9	9	54	14	2
IMSA GTD	12	**7**	18	14	2
IMSA GTD PRO	12	**9**	33	15	2
WEC Hypercar	2	2	39	**1**	1
ELMS LMP2	**2**	3	45	4	2
ELMS LMP2 Pro/Am	3	**2**	45	4	2

A rule of thumb sits closer to practice than the exact optimiser in 5 of 7 classes, ties it in 1 and loses in 1. In 3 of those classes the winning rule is B1, which uses no fitted quantity whatsoever — only the race length and the number of stops the tank forces. In the two it does not win, it ties the optimiser rather than losing to it.

That comparison is against the audit as first run. Held to each team's next stop, the optimiser is closer than every rule in 5 classes and ties the best rule in 2, the last column of Table 2. It is not a like-for-like contest: the replay knows the team's second stop and the rules do not, and in WEC Hypercar and ELMS the replay was shown above to be uninformative. What it does show is that the rules' advantage over the first-run optimiser came from the question that optimiser was asked, not from information it lacked.

Closer to what teams did is not *better*: every number here scores agreement with practice, never which plan was faster.

8 Result 4: which car counts as the rival is under-determined, and here it matters little

The replay above pairs each audited car with its nearest neighbour on classified position, which Thraves objected chooses the car ahead on some decisions and the car behind on others while the decision order is fixed, and which leaves 271 of 357 decisions as ties between an equidistant neighbour in front and one behind. He proposed a criterion in its place: estimate for each car the total race time it would set under its own optimal strategy in the absence of competition, and take as rival the car whose estimated time is closest.

I implement it with the same exact dynamic program the audit already uses, scoring each car from its own median green lap, the circuit-compound degradation posterior for the tyre it is running, and the circuit's measured pit loss. The compound term matters for the reason he gave: two cars at identical raw pace reach their optima on different laps if one is on a faster-wearing tyre. Formula 1 has not refuelled since 2010, so the fuel cap is the race distance and never binds, and what the program trades off is degradation against stop cost. The median gap in estimated standalone race time between an audited car and the rival this picks is 4.0 s over a full race.

Table 3 runs 4 rules against the same decisions.

Table 3: Rival-selection rules on the same Formula 1 decisions. Errors are signed in laps, positive meaning later than the team stopped. *Closed* is the single-car absolute error minus the cover-aware one, so a positive value would mean modelling the rival moved the recommendation toward the team's choice. The position-ahead rule scores fewer decisions because a race leader has no car ahead of it.

rule	scored	cover-aware	closed (median)	closed (mean)	improved
nearest on position	357	+11	-1	-0.96	63
within one pit loss	357	+11	-1	-1.06	55
car ahead on the clock	271	+11	-1	-0.75	56
nearest standalone time	357	+10	+0	-0.01	99

Two things come out of it and they point opposite ways.

The choice of rival is consequential. His rule and the published one name the same car on 20% of decisions, and all 4 rules agree on 17 of the 271 decisions where all of them apply. Four defensible readings of the word "rival" disagree about which car to model four times out of five.

It does not move Result 3. The cover-aware recommendation stays later than the real stop under every rule, by a median of +10 laps at best against 12 for the single-car model it is meant to improve on. What his rule does change is the penalty: under the published selection the

cover-aware model sits a median -1 lap further from the real stop than the single-car one, and under his it is level at +0, improving 99 decisions against 63. The defect he identified is therefore real, and correcting it does not produce the agreement the adversarial model was built to find.

He also named the limit of his own proposal, and I state it as he put it: at the extreme the choice of rival becomes endogenous, since each car's strategy depends on the others. The standalone time this heuristic sorts on is computed with no competition in it, which is what makes it computable and equally what makes it an approximation. I do not resolve this. A selection rule that accounted for the interaction would have to solve the joint problem the selection is a step toward, and nothing here does that.

A second approximation is mine rather than his. The standalone time is the whole race from the start rather than the remaining race from the decision lap, so it pairs cars by capability instead of by where they have got to. His formulation is the former and it is what I implement; the latter would answer a different question.

What this does and does not claim. How far four rules disagree depends on how different they are, and four near-restatements of one criterion would of course agree. The obvious objection is therefore that the rules were chosen to diverge. They were not, and the provenance of two of them is what answers it.

The car-ahead rule is not a reading I selected. It is the pairing Salminen [2026] defines his defences on: his leader and chaser are whichever two cars are adjacent on the clock, so testing his model at all forces that rule and no other. The standalone-time rule was proposed to me by Charles Thraves, in correspondence, as a criterion he thought the audit should use instead of mine; he is an author of the Stackelberg treatment (Aguad and Thraves, 2024; Carrasco Heine and Thraves, 2022) whose framing this project's adversarial component reuses. Neither rule was constructed here, neither was constructed against the other, and neither was introduced because it disagreed — the disagreement is what came back.

The disagreement is also bounded. No decision produces four distinct cars, and 148 of 271 produce exactly two. The claim is that a rival is not determined by the timing data together with a defensible criterion, which is weaker than saying it is undefined and strong enough to matter for any model that assumes one.

The game-theoretic branch of this literature is built on a two-car abstraction. Aguad and Thraves [2024] solve a Stackelberg game between two drivers, de Vries et al. [2026] pose a two-

agent problem per lap, and Salminen [2026] derives his equilibrium for a leader and a chaser. Each is internally consistent and none of them has occasion to say which two cars those are in a real race, because the question does not arise inside the abstraction. Measured against real decisions, the identification is the part that gives way.

9 Threats to validity

Estimators built here and withdrawn. A piecewise race-time basis intended to absorb track evolution was validated on synthetic races and then withdrawn twice on real ones: real fields sit close to the degenerate boundary where tyre age and race time share most of their variance after fixed effects. Formula 1's median identifiability is 0.691, well outside the range the basis was shown to work in. I report the omitted variable as diagnosed and unfixed rather than applying a correction that fails silently.

Result 1 bounds reuse, not refitting. Giona Fieni (ETH Zürich) stated the scope limit on this result more sharply than I had. A primarily model-based policy treats fitted degradation curves as placeholders and retunes them for each car and circuit, so it never depends on a slope transferring to a season it has not seen. Result 1 measures the stability of reused data-driven parameters. That bounds approaches which carry fitted parameters forward, and says nothing against a design that refits as a matter of course; on that design's own terms it is an argument for refitting.

The concession stops there. If measured behaviour were stable once the physics is fixed, the near-specification control field should have been the one that transferred: one chassis, one engine, no Balance of Performance, a single tyre supplier. It fails as the performance-balanced fields do. So either lap time carries large effects that are not the tyre, or the retuning step is doing more than retuning. I read it as the first, and the track evolution withdrawn above is my candidate for part of what it is.

The audit was not a fair comparison, and most of Result 3 was that. The engine plans one stop and runs to the flag; a pit wall chooses a lap inside a strategy it has already committed to. Section 7.4 holds the team's next stop fixed and most of the gap disappears. Three limits remain. Holding the second stop fixed gives the engine the team's plan in hindsight; restricting to green-flag second stops answers the part of that objection that concerns opportunistic stops,

not the part that concerns a plan revised during the race. The engine still represents neither the tyre allocation nor traffic, so the one-stop residual of +4 laps may yet be either; and that residual is partly the audit's asymmetric window, as it shrinks when the planner is asked earlier (Section 7.4). And in WEC Hypercar and ELMS the replay cannot discriminate, because the fuel range leaves the engine too few laps to choose from. Section 7.5 prices the residual from the races themselves, and the price is small; pricing it does not explain it, and that measurement is an association between cars, open to every reason a team had for its lap.

The endurance tyre clock. Endurance teams often double-stint tyres, changing them at every other fuel stop, and the decision they face is often whether to change tyres at a fuel stop rather than whether to add one. The endurance model counts tyre age from the last stop and carries one net slope with no compound, so it neither measures the real age of a double-stinted set nor separates the compounds Hypercar teams choose between. Its tyre-limited race-seasons answer a question endurance teams rarely ask, and its replay, as Section 7.4 shows, cannot test the tyre model at all.

What was decided after seeing the data. The committed replay, the rival-selection correction and the design checks were all built after the first-run gap was seen. Each is documented with the result that prompted it, but the only real protection against a forking path is a test the analysis could not have seen. The protocol is frozen for one (Section 10).

Drivers manage tyres to the stint they were given. A driver told to make a set last a long stint drives it more gently, so the slope a fit measures depends partly on the strategy the team chose. A planner fitted on those slopes will tend to find long stints affordable, which pushes in the direction of the one-stop residual, and season-to-season changes in how teams manage tyres would show up as the instability of Result 1. The cross-source check in Section 7 separates tyre wear from fuel burn, not this. Public timing cannot remove it; it could be bounded by comparing stints of the same car with different planned lengths, which I have not done.

Unresolved slopes, carried forward anyway. Of the 73 Formula 1 degradation coefficients, 16 have a cluster-robust interval containing zero, and 75% of compound pairs within a circuit have intervals that overlap each other. Every one of them is carried forward regardless. Screening on the resolved ones would select on the dependent variable — circuit-classes with tight slopes

are exactly the ones likeliest to transfer — and the transfer result would then be partly an artefact of the screen. The cost is that the weak transfer reported in Section 4 is a mixture of genuine instability and of parameters the data never pinned down. Section 4.1 separates those two for Formula 1 by measuring what a stable slope could have scored against the same noise, and the mixture turns out to matter: most circuits cannot clear the quoted threshold even with a slope that does not move. The endurance classes are not separated this way, and their counts should be read with that in mind.

Constants resting on very little. The cheap-stop edge of 22.5 s is a maximum set by one race; drop it and the edge falls to 13.2 s, 41%. The second-best transfer score, +0.497 at Lime Rock GTD PRO, is a mean over 2 folds running from +0.357 to +0.638 — a spread of 0.281, wider than most circuit-classes' entire score, and it should not be read beside Lime Rock GTD's +0.573, which averages 3 folds spanning 0.024. Same circuit, same protocol, an order of magnitude more stable. The instability is therefore not a property of the circuit alone. And the top of the track-position range rests on 3 races at a circuit first run in 2023. The rules these support are not thin — 150 race-seasons sit above the edge with 0 tyre-limited among them — but the constants are, and they are quoted here as orders of magnitude. A full accounting is generated into `reports/cross_series/thin_evidence.md`.

Compound is unavailable in endurance. The endurance degradation models are net-slope models. Comparisons between the Formula 1 per-compound coefficients and endurance net slopes are made only where the quantity is genuinely comparable.

Neutralisation regimes are not poolable. IMSA records no Safety Car in any scoped race and a Full Course Yellow in nearly all of them; WEC and ELMS are Safety-Car championships at very different rates. Any pooled endurance neutralisation model would sit between these and describe none of them, so they are never merged.

Some layers cannot be rebuilt from a clone. The breadth-degradation and reliability layers read a third-party export that is not redistributed here. Their outputs are committed, so every downstream consumer and every test runs offline; what is blocked is rebuilding them from source. Section 10 names them, and names the one whose blockage reaches a published result.

10 Reproducibility

Every artifact derived from committed data is regenerated by continuous integration, which fails on any difference between the working tree and the repository, including files a generator wrote and nobody committed.

The state these results were computed from is archived on Zenodo and, through it, in Software Heritage and OpenAIRE. The concept identifier `https://doi.org/10.5281/zenodo.22726130` resolves to the newest version of the deposit. The release archived alongside this paper carries its own frozen identifier, recorded in the repository's `CITATION.cff`, for anyone who needs the exact state a figure came from rather than the current one.

The blocked layers, named. 3 layers read third-party exports this repository does not redistribute, so those layers cannot be regenerated from a clone: the Formula 1 breadth layer and the Formula 1 and WEC reliability tables. Their outputs are committed, so everything downstream of them runs offline, and the rest of the project — Formula 1 core, WEC, IMSA, ELMS, the simulator, the audits and every cross-series layer — reproduces from a clone with no external download. The Pirelli nominations and the practice laps of Section 4.2 are committed with the scripts that collected them, so they too are read offline.

One of the three is load-bearing. The cross-source slope check that rejected one of the candidate explanations for the gap in Result 3 rests on the breadth layer, whose input a reader cannot obtain from this repository. Reproducing that rejection requires the external source, which is named in the repository along with what each blocked layer reads and writes.

A prospective test, frozen before its data exist. The protocol of the committed replay — the planner, its seeds, the lookback, the selection of cars and the horizon — is frozen in `paper/prospective_protocol.md`, committed before the Formula 1 races it will be scored on had been run. It states in advance the hypothesis to be tested on every race from the São Paulo Grand Prix of the current season onward: that the committed median gap on those races falls inside the interval reported here, [+1.0, +4.0]. The repository's history dates the file, and the result will be reported against it whichever way it falls.

Every number in this manuscript is a macro generated by `scripts/make_paper_numbers.py` from the committed artifacts; the source of this paper contains no digits. A separate test recomputes each published headline from its artifact and asserts the formatted result appears

in the document that publishes it, which is stricter than recomputing alone.

11 Conclusion

What this paper offers is a protocol and what came back from it. Degradation slopes rarely transfer, and where they do it is a property of the circuit-class rather than of the championship; Balance of Performance is not needed for the instability, and the effect of car type cannot be separated here from the circuits each type races on. Whether an extra stop can ever pay is decided by the cost of the stop, which separates the two regimes at an AUC of 0.977 against 0.865 for the tyre. A time-optimal planner first appeared to disagree with professional practice by a median of +10 laps in Formula 1. Four explanations failed against that figure; the fifth was the audit itself: asked to plan the stint the team would actually run, the planner lands +2 laps from the real stop rather than +9.5 (Section 7.4), across lookbacks and for the whole field. In endurance, where one net slope leaves the timing independent of the tyre, the replay cannot test the tyre model at all (Section 7.4). Simpler rules landed closer to practice than the first-run audit in 5 of 7 classes. Two repairs a team would reach for first, recoding each circuit's tyres by their real compound and measuring degradation on Friday, both leave transfer where it was or worse (Section 4.2), past-only fitting costs a little, and a circuit's own slope does not beat one common to the category (Section 4.3). Where the laps carry signal, a slope from other seasons recovers a median 0.71 of what a stable one could; the larger lesson is how few circuits carry enough signal to ask.

And the rival that a competition-aware model should answer is under-determined: 4 defensible rules name the same car on 17 of 271 decisions, which constrains every model built on a two-car abstraction.

What remains of the planner–team gap is +4 laps in one-stop races, where the planner and the team were answering the same question. The planner prices it at tenths of a second per lap of distance, and the races cannot decide whether that price is right (Section 7.5). The two mechanisms rejected against the first-run audit stay rejected on it (Section 7.4), and part of it is the asymmetric window the audit gives the planner: asked fifteen laps ahead, it falls to +2 (Section 7.4). What remains reads as a parameter: teams stop as if their worn tyre were degrading faster, relative to the new one, than the fit says. It is smaller than the gap this paper set out to explain, and it is the one I would most like someone else to explain.

Data and code availability

Every quantity in this paper is derived from committed artifacts by scripts in a public repository, which also contains the data, the figures and the tests: `https://github.com/mohammedmedjadj/Motorsport-Strategy-Lab`. The state the results were computed from is archived and citable at `https://doi.org/10.5281/zenodo.22726130`.

Acknowledgements

Several researchers replied to cold emails about this work. Three of those replies changed what the repository contains, and what they changed is set out here rather than thanked in general terms.

Charles Thraves (Universidad de Chile) identified the rival-selection defect in the adversarial replay: minimising absolute classified-position distance picks the car ahead on some decisions and the car behind on others, and with equidistant neighbours the choice is incidental. He then proposed the standalone-optimal-time heuristic tested in Section 8, confirmed that compound and tyre-age differences in lap time are enough to move the relative timing, and named the endogeneity that bounds his own proposal.

Wytze de Vries (TU Eindhoven) raised two measurement objections to Result 2: that a class-level median pit loss pools two-tyre stops with full service including a driver change, and that part of the correlation may run backwards, since tyre-limited races carry less fuel and so stop faster. Both were tested; the first holds and does not disturb the class ordering, and the second has the sign he predicted at a magnitude too small to account for the result.

Jorn van Kampen (TU Eindhoven) described the same structure from electric endurance racing, where the charging stop is long enough that a tyre change is treated as negligible and is not optimised separately.

Giona Fieni (ETH Zürich) drew the line between a result about reused data-driven parameters and a result about model-based policy, which is the scope limit now stated in Section 9; and held that once the physics is known the behaviour follows, which is the position the near-specification control field is set against there.

None of them is responsible for what this manuscript says.

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